- sectoral specialisation
- close inter-firm collaboration
- inter-firm competition based on innovation
- a socio-cultural identity, which facilitates trust
- multi-skilled workforce
- active self-help organisations, and
- supportive regional and municipal governments.

Firms are therefore expected to collaborate and compete with one another at the same time. By collaborating, they can expand their capacity and also learn from one another. Through competition, they are forced to become more efficient.

How Do Clusters Originate?

Clusters may arise due to many factors. Certain clusters evolve over a long time in history when artisans settle in one locality and evolve over centuries. Handloom weaving clusters are one examples of this development. Or else, in some sectors, when a large firm is established, a cluster of firms may emerge to take care of its input and service requirements. At times, governments may decide to encourage manufacturing using raw materials from a region, which may also lead to emergence of clusters.

5.4 Historical Development of Industrialisation in Tamil Nadu

There is lot of evidence for presence of industrial activities such as textiles, ship-building, iron and steel making and pottery in precolonial Tamil Nadu. Given the vast coastline, the region has been involved in trade with both South-East and West Asia for several centuries. Colonial policies also contributed to the decline of the handloom weaving industry due to competition from machine-made imports from England. But some industries also developed during the colonial period.

Industrialisation in the Colonial Period

There are two sets of factors that have contributed to the process. The introduction of cotton cultivation in western and southern Tamil Nadu by the colonial government led to the emergence of a large-scale textile sector in these parts.

Second, increase in trade during this period led to industrial development around two of the most active ports in the region, Chennai and Tuticorin. Match factories too emerged during the colonial period in the Sivakasi region, which later on became a major centre for fireworks production and printing. Port-related activity too contributed to the growth of the region. Leather production was also taking place in Dindigul, Vellore and Ambur areas.

In Western Tamil Nadu, the emergence of textiles industries also led to demand and starting of textile machinery industry in the region. This textile machinery industry in turn led to the rise of a number of small workshops for repair and producers of machinery components. Another major development in the western region is the introduction of electricity from hydro-electric power in 1930s. Availability of electricity allowed for use of oil engines for drawing ground water. This led to both expansion of agriculture as well as increase in demand for oil engines. In turn, it led to emergence of workshops for servicing engines and also for addressing the demand for spare parts. Foundries began to be set up and agricultural machinery began to be produced.

Post-Independence to early 1990s

Soon after independence, several large enterprises were set up by both the central and state governments in different segments such as the Integral Coach Factory in Chennai to make railway coaches and the Bharat Heavy Electricals Limited (BHEL) in Tiruchirapalli manufacture to boilers and turbines. BHEL in turn led to the emergence of an industrial cluster of several small firms catering to its

27_Economics_Unit_5_EM.indd 330 1/4/2022 8:54:08 PM